# MLJ Contribution Information Sheet

**Manuscript:** *CascadeRAG: Abstention-Triggered Escalation that Outperforms Costlier Retrieval-Augmented Generation Architectures at a Fraction of their Cost*
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## 1. What is the main claim of the paper? Why is this an important contribution to the machine learning literature?

**Main claim.** Among the architectural refinements proposed for retrieval-augmented generation (RAG), none is universally safe, and the design that wins a controlled comparison is a cascade that spends extra computation only where a free, posterior signal says it is needed. The paper makes three concrete claims. (i) Under one controlled stack, the strongest single refinement, cross-encoder reranking, gains +0.051 Exact Match on one question set and loses 0.100 on another, a reversal that single-benchmark evaluations do not reveal. (ii) A per-question diagnosis of *why* retrieval fails (five causes) predicts which refinement pays off on which question set. (iii) CascadeRAG, a novel architecture built from that diagnosis, fuses dense, lexical, and cross-encoder rankings and escalates a question to a more expensive strategy only when the reader itself, having seen the evidence, declines to answer. This abstention trigger is a reject option with provably zero selective risk under Exact Match, so escalation can never discard a correct answer. CascadeRAG wins 41 of 57 paired (baseline, question set) comparisons, loses 1, and ties 15, at 1.16–1.58 generator calls per question against 2.3–3.0 for iterative and corrective methods.

**Why it matters.** RAG is now the standard way of grounding large language models in a document collection, yet its architectures are compared in the literature as point scores on incompatible stacks. The paper supplies two things the machine learning literature lacks here: a research-methodology contribution, namely a comparison protocol with paired significance tests under multiplicity control, budget-matched controls, abstention accounted for separately from error, and a measured corpus-imposed recall ceiling, under which most published-style gains shrink and one reverses; and a learning-systems contribution that connects RAG design to selective prediction, showing that a training-free, posterior deferral signal dominates both prior routing and confidence-threshold deferral. Two documented negative results (neighbour-chunk expansion adds nothing once context size is controlled; answer-token confidence adds nothing over abstention) save practitioners from two widely assumed improvements.

## 2. What is the evidence you provide to support your claim? Be precise.

Thirteen RAG architectures are evaluated on a single stack: one 21-million-passage Wikipedia corpus, one embedding model, one vector database, one generator (Qwen2.5-7B-Instruct), and one answer prompt, over five question sets (NaturalQuestions, TriviaQA, HotpotQA, 2WikiMultihopQA, MuSiQue) totalling more than 37,000 questions. Differences are tested with paired statistics (McNemar's test on Exact Match, the Wilcoxon signed-rank test on continuous metrics) with Holm-adjusted p-values, and every difference carries a paired bootstrap 95% confidence interval. Budget-matched controls separate a better architecture from more text in the context. The main results are: (a) cross-encoder reranking gains +0.051 Exact Match on HotpotQA and loses 0.100 on NaturalQuestions; (b) the distribution of five retrieval failure causes, measured per question, explains these reversals; (c) CascadeRAG wins 41 of 57 paired comparisons against every baseline on every question set, loses 1, and ties 15, at 1.16–1.58 generator calls per question; (d) the losslessness of the abstention trigger holds with zero exceptions over all 887,917 (question, architecture) pairs measured; (e) the result replicates under a second, unrelated generator (Llama-3.1-8B-Instruct); (f) ablations isolate what the fused first stage and the escalation each add, and show that a confidence threshold adds nothing. Every number in the manuscript is generated by the benchmark code from the per-question logs and imported as a named macro; the code, the logs, and the manuscript pipeline are publicly released.

## 3. What papers by other authors make the most closely related contributions, and how is your paper related to them?

- Jin, Zhu, Yang, Zhang, and Dou (2024) (FlashRAG) standardise datasets, corpora, and reimplementations of a dozen RAG methods, and are the source of our corpus and question sets. We add paired significance testing with multiplicity control, the corpus-imposed ceiling, abstention accounting, and a per-question failure diagnosis, and we study a different set of architectures under one stack.
- Jeong, Baek, Cho, Hwang, and Park (2024) (Adaptive-RAG), Asai, Wu, Wang, Sil, and Hajishirzi (2024) (Self-RAG), and Yan, Gu, Zhu, and Ling (2024) (CRAG) all decide how much work a question needs, from the question alone or with a trained component. CascadeRAG decides *after* the reader has seen the evidence (a posterior signal) and requires no trained component; Adaptive-RAG and CRAG are rows in our comparison.
- Chow (1970) and Geifman and El-Yaniv (2017) formalise the reject option and selective prediction; we show that the reader's abstention is a reject option with zero selective risk under Exact Match and exploit it as a deferral trigger.
- Viola and Jones (2001) and L. Wang, Lin, and Metzler (2011) introduce cascades in vision and in cascade ranking for retrieval; we apply the idea at the level of whole RAG strategies.
- X. Wang et al. (2024) tune RAG components in isolation; our controlled comparison shows that the gains of such components depend on the question distribution and can reverse.

## 4. Have you published parts of your paper before, for instance in a conference?

No. No part of this manuscript has been published before, in a conference or elsewhere, and the manuscript is not under review at any other venue.

## References


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